

1. In the game *Minecraft*, suppose the duration of a thunderstorm follows a normal distribution with a mean of μ minutes and a standard deviation of 1.5 minutes.
- (a) 150 random samples of thunderstorms are observed, and it is found the sample mean of the duration of a thunderstorm is 7.8 minutes. Another random sample of 19 observations is selected. The sum of minutes of thunderstorm in these 19 bottles is 215.8 minutes. Using these 169 combined samples, construct a 95% confidence interval for μ . (4 marks)
- (b) Suppose $\mu = 8$. Find the probability that a thunderstorm will last between 5.75 and 10.505 minutes. (2 marks)
- (c) During a thunderstorm, for each loaded chunk, every tick there is a 10^{-5} chance of a thunderstrike occurring. Given: 1 minute = 1200 ticks.
- (i) Denote the number of thunderstrikes in 1 chunk in 8 minutes by X. Using a **suitable approximation**, write down the probability distribution of X.
- (ii) If 16 chunks are loaded, what is the probability more than 2 chunks would have at least one thunderstrike, if the thunderstorm lasts for 8 minutes? (4 marks)
- (d) Using the trapezoidal rule with 5 sub-intervals, estimate the probability of exactly 2 thunderstrikes in 12 chunks, given that the rain lasts between 5.75 and 10.505 minutes. Hint: consider
- $$F(x|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt$$
- (6 marks)
- (e) Mario claims the probability of 2 thunderstrikes occurring when 12 chunks is loaded and the rain lasts between 5.75 and 10.505 minutes is higher than 0.2. Do you agree with Him? Justify your answer. (4 marks)

a) New sample mean

$$= \frac{7.8 \times 150 + 215.8}{169}$$

$$= 8.2 \text{ minutes} \quad \text{IA}$$

95% confidence interval

$$= \left(8.2 - (1.96) \left(\frac{1.5}{\sqrt{169}} \right), 8.2 + (1.96) \left(\frac{1.5}{\sqrt{169}} \right) \right) \quad \text{IM+IA}$$

$$= (7.9738, 8.4262) \quad \text{IA}$$

b) required probability

$$= P\left(\frac{5.75-8}{1.5} < Z < \frac{10.505-8}{1.5}\right) \quad \text{IM}$$

$$= 0.4332 + 0.4525$$

$$= 0.8857 \quad \text{IA}$$

$$\text{ci) } \lambda = 10^{-5} \times 1200 \times 8 = 0.096$$

$$\therefore X \sim \text{Po}(0.096) \quad \text{IA}$$

$$\text{iii) } \lambda = 0.096$$

P(at least one thunder strike in one chunk)

$$= 1 - e^{-0.096} \left(\frac{0.096^0}{0!} \right)$$

$$= 1 - e^{-0.096} \quad \text{IM, can be absorbed}$$

required probability

$$= 1 - \sum_{x=0}^2 C_{16}^x (1 - e^{-0.096})^x (e^{-0.096})^{16-x} \quad \text{IM}$$

$$= 0.175536566$$

$$= 0.1755 \quad // \quad 1A$$

d) let t be the time (in minutes) elapsed.

$$\lambda = 10^{-5} \times 1200 \times t \times 12 = 0.144t$$

required probability

$$= \left(\frac{e^{-0.144t} (0.144t)^2}{2!} \right) \left(P\left(\frac{5.75-8}{1.5} < Z < \frac{10.505-8}{1.5}\right) \right)^{-1} \cdot \int_{5.75}^{10.505} e^{-\frac{(t-8)^2}{4.5}} dt \left(\frac{1}{1.5\sqrt{2\pi}} \right) \quad 1M + 1M + 1M$$

$$= \frac{108}{15625 (\sqrt{2\pi}) (0.8857)} \int_{5.75}^{10.505} e^{-0.144t} t^2 e^{-\frac{(t-8)^2}{4.5}} dt$$

for poisson
for conditional probability
for correct integral

$$\text{let } f(t) = e^{-0.144t} t^2 e^{-\frac{(t-8)^2}{4.5}}$$

$$\int_{5.75}^{10.505} f(t) dt \approx \frac{1}{2} \left(\frac{4.755}{5} \right) \left(f(5.75) + f(10.505) + 2[f(6.701) + f(7.652) + f(8.603) + f(9.554)] \right)$$

$$\approx 65.9209561 \quad 1M$$

required probability

$$= \frac{108}{15625 (\sqrt{2\pi}) (0.8857)} (65.9209561)$$

$$= 0.205234618$$

$$= 0.2052 \quad // \quad 1A$$

$$e) f(t) = e^{-0.144t} t^2 e^{-\frac{(t-8)^2}{4.5}}$$

$$= t^2 e^{-0.144t - \frac{2}{9}(t-8)^2}$$

$$f'(t) = e^{-0.144t - \frac{2}{9}(t-8)^2} \left(t^2 \left(-0.144 - \frac{4}{9}(t-8) \right) + 2t \right) \quad 1M$$

Answers written in the margins will not be marked.

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$$f'(t) = e^{-0.144t - \frac{2}{9}(t-8)^2} \left(-\frac{4}{9}t^3 + \frac{3838}{1125}t^2 + 2t \right)$$

$$f''(t) = e^{-0.144t - \frac{2}{9}(t-8)^2} \left(-\frac{4}{3}t^2 + \frac{7676}{1125}t + 2 + \left(-\frac{4}{9}t^3 + \frac{3838}{1125}t^2 + 2t \right) \left(\frac{3838}{1125} - \frac{4}{9}t \right) \right)$$

$$= e^{-0.144t - \frac{2}{9}(t-8)^2} \left(-\frac{4}{3}t^2 + \frac{7676}{1125}t + 2 + \left(-\frac{8}{9} \left(\frac{3838}{1125} \right) t^3 + \left(\left(\frac{3838}{1125} \right)^2 - \frac{8}{9} \right) t^2 + \frac{7676}{1125}t + \frac{16}{81}t^4 \right) \right)$$

$$= e^{-0.144t - \frac{2}{9}(t-8)^2} \left(\frac{16}{81}t^4 - \frac{8}{9} \left(\frac{3838}{1125} \right) t^3 + \left(\left(\frac{3838}{1125} \right)^2 - \frac{20}{9} \right) t^2 + \frac{15352}{1125}t + 2 \right)$$

note that $f'(t) < 0$ for all $x \in [5.75, 10.505]$ 1A ft

$\therefore$ it is an underestimate

i.e. probability > 0.2052

$\therefore$ Mario's claim is correct. 1A ft

$$4 + 2 + (1 + 3) + 6 + 4 = 20$$

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Footnotes

- In reality, the values for the range of duration of a thunderstorm are uniformly distributed, from 3 to 13 minutes. Notably, by the principle of central limit theorem, answer obtained in (a) should be the same.
- Another information such as the probability of thunderstrike and tick conversion is the same as the real game.
- This question has been simplified; as in the real game factors such as sleeping and thunderstorm probability have not been mentioned.